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High conductance variable orifice valve

Abstract

Variable orifice valves comprising a first fixed plate, a second fixed plate and a movable plate between are described. The movable plate is connected to the first fixed plate and the second fixed plate by sealing elements. The movable plate is moved closer to or further from the first fixed plate by rotation of an actuator ring that rotates at least two rotary elements connected to the movable plate. A needle on the movable plate engages an opening in the valve to seal or open the valve to allow fluid flow. Methods of controlling flow of fluid through the variable orifice valve are also described.

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Background/Summary

TECHNICAL FIELD

(1) Embodiments of the disclosure generally relate to apparatus for controlling gas pressure or flow rate. In particular, embodiments of the disclosure relate to variable orifice valves that can operate at a wide conductance range and/or higher temperatures.

BACKGROUND

(2) Semiconductor manufacturing using many low-pressure processes in specially designed processing chambers. These processing chambers are frequently subjected to pumping and purging

processes to maintain suitable processing conditions. During formation of electronic devices using, for example, chemical vapor deposition (CVD) or atomic layer deposition (ALD) processes, the process chamber gas flow rate is controlled to meet the deposition and/or purge requirements of the process.

(3) In ALD/CVD processes, gas flow rates can be varied with varying valve conductance. With the changing of the gas flow rate, for example, precursor delivery (concentration), purge flow rate (dilution), processes can be optimized in terms of film thickness and uniformity. Currently available variable orifice valves are limited to low conductance ranges and cannot be operated at high temperatures, limiting the usefulness of these valves in the deposition processes.

(4) Accordingly, there is a need for apparatus and methods to provide a wide operating range of gas conductance at high temperatures.

SUMMARY

(5) One or more embodiments of the disclosure are directed to variable orifice valves including: a first fixed plate, a second fixed plate, a movable plate, a first sealing element, a second sealing element, and an actuator ring. The first fixed plate is located at a first end of the valve. The first fixed plate has a first side and a second side defining a first fixed plate thickness. The first fixed plate has an opening extending through the first fixed plate thickness.

(6) A second fixed plate is at a second end of the valve. The second fixed plate has a first side and a second side defining a second fixed plate thickness. The second fixed plate has an opening extending through the second fixed plate thickness. The first side of the second fixed plate faces and is spaced from the second side of the first fixed plate.

(7) The movable plate is positioned between the first fixed plate and the second fixed plate. The movable plate has a first side and a second side defining a movable plate thickness. The first side of the movable plate faces the second side of the first fixed plate and the second side of the movable plate faces the first side of the second fixed plate.

(8) The movable plate has a needle extending from the first side of the movable plate. The needle has a needle end sized to fit within the opening in the first fixed plate. The movable plate has at least one opening extending through the movable plate thickness adjacent to the needle.

(9) The first sealing element is between the first fixed plate and the movable plate. The second sealing element is between the second fixed plate and the movable plate. The actuator ring is adjacent to the first fixed plate.

(10) The actuator ring has a plurality of inwardly directed engagement elements on an inside surface thereof. At least two rotary elements are within the actuator ring. Each of the rotary elements has a plurality of outwardly directed engagement elements on an outside surface thereof. Each of the outwardly directed engagement elements is configured to cooperatively interact with the inwardly directed engagement elements of the actuator ring. At least one connector is between the first fixed plate and the movable plate. The at least one connector is configured to move the movable plate upon rotation of the actuator ring.

(11) Additional embodiments of the disclosure are directed to variable orifice valves including: a first fixed plate, a second fixed plate, a movable plate, a first sealing element, a second sealing element, an actuator ring, a first conduit and a second conduit. The first fixed plate is located at a first end of the valve. The first fixed plate has a first side and a second side defining a first fixed plate thickness. The first fixed plate has an opening extending through the first fixed plate thickness.

(12) The second fixed plate is located at a second end of the valve. The second fixed plate has a first side and a second side defining a second fixed plate thickness. The second fixed plate has an opening extending through the second fixed plate thickness. The first side of the second fixed plate faces and is spaced from the second side of the first fixed plate.

(13) The movable plate is positioned between the first fixed plate and the second fixed plate. The movable plate has a first side and a second side defining a movable plate thickness. The first side of

the movable plate faces the second side of the first fixed plate and the second side of the movable plate faces the first side of the second fixed plate. The movable plate has a needle extending from the first side of the movable plate. The needle has a needle end sized to fit within the opening in the first fixed plate. The movable plate has at least one opening extending through the movable plate thickness adjacent to the needle. The needle end has a conical shape with a maximum width sized to contact the second side of the first fixed plate at the opening to form a seal.

(14) The first sealing element is between the first fixed plate and the movable plate. The first sealing element includes a bellows. The second sealing element is between the second fixed plate and the movable plate. The second sealing element includes a bellows.

(15) The actuator ring is adjacent to the first fixed plate. The actuator ring has a plurality of inwardly directed engagement elements on an inside surface thereof. At least three rotary elements are within the actuator ring. Each of the rotary elements has a plurality of outwardly directed engagement elements on an outside surface thereof. Each of the outwardly directed engagement elements are configured to cooperatively interact with the inwardly directed engagement elements of the actuator ring. At least one connector is between the first fixed plate and the movable plate. The at least one connector is configured to move the movable plate upon rotation of the actuator ring.

(16) The first conduit extends from the first side of the first fixed plate. The first conduit has a threaded end and outwardly facing engagement elements configured to cooperatively interact with the outwardly directed engagement elements of the at least three rotary elements. The second conduit extends from the second side of the second fixed plate. The second conduit has a threaded end.

(17) Movement of the actuator ring around a central axis causes each of the rotary elements to rotate around a rotary element axis while remaining in a fixed location relative to the first conduit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. 1 shows a variable orifice valve according to one or more embodiment of the disclosure;

(3) FIG. 2 shows a cross-sectional view of variable orifice valve according to one or more embodiment of the disclosure;

(4) FIG. 3 shows a schematic cross-sectional view of a movable plate for use with a variable orifice valve according to one or more embodiment of the disclosure;

(5) FIG. 4 shows a portion of a needle end of a needle that is attached to a movable plate according to one or more embodiment of the disclosure;

(6) FIG. 5 shows a perspective view of an actuator ring with three rotary elements and indicia according to one or more embodiment of the disclosure;

(7) FIG. 6A shows a portion of a schematic cross-sectional view of a needle engaging an actuator plate fully sealing a variable orifice valve according to one or more embodiment of the disclosure;

(8) FIG. 6B shows a portion of a schematic cross-sectional view of a needle disengaged from the actuator plate fully opening a variable orifice valve according to one or more embodiment of the disclosure;

(9) FIG. 7A shows a portion of a needle with a convex needle end according to one or more

embodiment of the disclosure;

(10) FIG. 7B shows a portion of a needle with a convex end according to one or more embodiment of the disclosure; and

(11) FIG. 8 shows a motor driven variable orifice valve according to one or more embodiment of the disclosure.

(12) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(13) Before describing several exemplary embodiments of the disclosure, it is to be understood that the disclosure is not limited to the details of construction or process steps set forth in the following description. The disclosure is capable of other embodiments and of being practiced or being carried out in various ways.

(14) As used in this specification and the appended claims, the term “substrate” refers to a surface, or portion of a surface, upon which a process acts. It will also be understood by those skilled in the art that reference to a substrate can also refer to only a portion of the substrate, unless the context clearly indicates otherwise. Additionally, reference to depositing on a substrate can mean both a bare substrate and a substrate with one or more films or features deposited or formed thereon.

(15) Embodiments of the disclosure are directed to variable orifice valves that can be manually or automatically operated that allow for a wide range of gas conductance and can operate at high temperatures. Some embodiments operate at temperatures greater than or equal to 50

°C, 100°C, 150°C, 200°C, 250°C, or 300°C for periods greater than or equal to 1 hour, 2 hours, 3 hours, 4 hours or 5 hours.

(16) One or more embodiments of the variable orifice valve comprises a bellows with moving datum plate (also referred to as a movable plate). The movable plate can be operated by movement of a circular dial with screw (or similar) connections. A needle on the movable plate is moved towards fixed datum plate (also referred to as a fixed plate) by actuation of the dial. The dial of some embodiments acts as an annular gear assembly and internally threaded to operate multiple screws at the same time. The bellows allows for the flexibility of the moving plate to vary the valve conductance without system leakage. In some embodiments, the dial is manually or automatically controlled using a motor. The valve conductance can be varied using a wide range of openings designed as required for a particular process.

(17) Some embodiments of the disclosure advantageously provide variable orifice valves that are compact and require less space for installation than current valves. Some embodiments can be operated at a wide temperature range. Some embodiments provide valve conductance that can be optimized over a wide range as required by particular process conditions. Some embodiments are advantageously easy to operate compared to available need valves using dial rotation to actuate the valve.

(18) With reference to the Figures, one or more embodiments of the disclosure is directed to a variable orifice valve **100**. The variable orifice valve **100** has a first end **101** and a second end **102** which can be inserted into a gas flow path to control the flow conductance.

(19) The variable orifice valve **100** comprises a first fixed plate **120**, a second fixed plate **140** and a movable plate **160**. The movable plate **160** is positioned between the first fixed plate **120** and the second fixed plate **140**.

(20) The first fixed plate **120** is located at or adjacent to the first end **101** of the variable orifice valve **100**. The first fixed plate **120** has a first side **122** and second side **124** that define a first fixed plate thickness $T_{sub.F1}$. The first fixed plate thickness $T_{sub.F1}$ can be any suitable thickness including, but not limited to, a thickness in the range of 0.5 mm to 50 mm, or in the range of 1 mm

to 10 mm, or in the range of 2 mm to 5 mm.

(21) The first fixed plate **120** has an opening **126** extending through the first fixed plate thickness T.sub.F1. The opening **126** in the first fixed plate **120** can be any suitable diameter depending on the target conductance of the gas flowing through the variable orifice valve **100**. In some embodiments, the opening **126** has a diameter in the range of 0.25 inches to 5 inches, or in the range of 0.5 inches to 4 inches, or in the range of 1 inch to 2 inches.

(22) The second fixed plate **140** is located at or adjacent to the second end **102** of the variable orifice valve **100**. The second fixed plate **140** has a first side **142** and second side **144** that define a second fixed plate thickness T.sub.F2. The first side **142** of the second fixed plate **140** faces the second side **124** of the first fixed plate **120**. The second fixed plate thickness T.sub.F2 can be any suitable thickness including, but not limited to, a thickness in the range of 0.5 mm to 50 mm, or in the range of 1 mm to 10 mm, or in the range of 2 mm to 5 mm.

(23) The second fixed plate **140** has an opening **146** extending through the second fixed plate thickness T.sub.F2. The opening **146** in the second fixed plate **140** can be any suitable diameter depending on the target conductance of the gas flowing through the variable orifice valve **100**. In some embodiments, the opening **146** has a diameter in the range of 0.25 inches to 5 inches, or in the range of 0.5 inches to 4 inches, or in the range of 1 inch to 2 inches.

(24) With reference to FIGS. 1 through 3, the movable plate **160** is positioned between the first fixed plate **120** and the second fixed plate **140**. The movable plate has a first side **162** and a second side **164** defining a movable plate thickness T.sub.M. The first side **162** of the movable plate **160** faces the second side **124** of the first fixed plate **120** and the second side **164** of the movable plate **160** faces the first side **142** of the second fixed plate **140**. The movable plate **160** is positioned so that there is a gap **125** between the first side **162** of the movable plate **160** and the second side **124** of the first fixed plate **120**, and there is a gap **145** between the second side **164** of the movable plate **160** and the first side **142** of the second fixed plate **140**.

(25) The movable plate **160** has a needle **170** extending from the first side **162** of the movable plate **160**. The needle has a needle end **172** sized to fit within the opening **126** in the first fixed plate **120**. The movable plate **160** has at least one opening **166** extending through the movable plate thickness T.sub.M adjacent the needle **170**. The at least one opening **166** adjacent the needle **170** allows for a flow of fluid (e.g., a gas or liquid) to pass through the movable plate **160**. There can be any suitable number of openings **166**. For example, in some embodiments, there are four, five, six, seven, eight, nine, ten, eleven or twelve openings **166**.

(26) A first sealing element **180** is located between the first fixed plate **120** and the movable plate **160**. The first sealing element **180** creates a fluid tight seal between the first fixed plate **120** and the movable plate **160** during movement of the movable plate **160** closer to or further from the first fixed plate **120**. The first sealing element **180** can be any suitable component known to the skilled artisan that can form a fluid-tight seal. In some embodiments, the first sealing element **180** comprises a bellows.

(27) A second sealing element **185** is located between the second fixed plate **140** and the movable plate **160**. The second sealing element **185** creates a fluid tight seal between the movable plate **160** and the second fixed plate **140** during movement of the movable plate **160** closer to or further from the second fixed plate **140**. The second sealing element **185** can be any suitable component known to the skilled artisan that can form a fluid-tight seal. In some embodiments, the second sealing element **185** comprises a bellows.

(28) In some embodiments, as shown in the Figures, the movable plate **160** further comprises at least one outer opening **168**. The at least one outer opening **168** is located in the outer portion of the movable plate **160** and is outside of the first sealing element **180** and the second sealing element **185**.

(29) In some embodiments, the first sealing element **180** and/or the second sealing element **185** comprise a high temperature material. In some embodiments, the first sealing element **180** and/or

the second sealing element **185** comprise bellows made of, or lined with, a high temperature material.

(30) The variable orifice valve **100** further comprises an actuator ring **190** located adjacent to the first fixed plate **120**. The actuator ring **190** of some embodiments has a plurality of inwardly directed engagement elements **192** on an inside surface **191** thereof. The plurality of inwardly directed engagement elements **192** of some embodiments are gear teeth.

(31) At least two rotary elements **110** are located within the actuator ring **190**. Each of the at least two rotary elements **110** has a plurality of outwardly directed engagement elements **112** on an outside surface **111** thereof. Each of the plurality of outwardly directed engagement elements **112** are configured to cooperatively interact with the plurality of inwardly directed engagement elements **192** of the actuator ring **190**. For example, in some embodiments, the plurality of inwardly directed engagement elements **192** of the actuator ring **190** are gear teeth and the plurality of outwardly directed engagement elements **112** of the at least two rotary elements **110** are complementary gear teeth configured to mesh with the gear teeth of the actuator ring **190**.

(32) In use, movement of the actuator ring **190** around a central axis **105** causes the plurality of inwardly directed engagement elements **192** to rotate around the central axis **105**. The plurality of inwardly directed engagement elements **192** meshing with the plurality of outwardly directed engagement elements **112** of the at least two rotary elements **110** results in rotation of each of the at least two rotary elements **110** around a rotary element axis **113**.

(33) In some embodiments, the at least two rotary elements **110** are positioned on an actuator plate **195**. The actuator plate **195** has a first side **193** and a second side **194** defining a thickness $T_{sub.A}$ of the actuator plate **195**, as shown in FIG. 6A. The actuator plate **195** of some embodiments keeps the at least two rotary elements **110** stationary so that the rotary element axis **113** of each at least two rotary elements **110** remains in a fixed location relative to the first conduit. Stated differently, in some embodiments, the actuator plate **195** keeps the at least two rotary elements **110** in a fixed location relative to the central axis **105**, without rotating around the central axis **105**.

(34) The variable orifice valve **100** can have any suitable number of rotary elements **110**. For example, the drawings illustrate variable orifice valve **100** with three rotary elements **110**. In some embodiments, there are two, three, four, five, or six rotary elements **110** arranged symmetrically around the central axis **105** of the variable orifice valve **100**.

(35) Actuation of the actuator ring **190** causes rotation of the at least two rotary elements **110** around the rotary element axis **113**. This rotary movement is translated into movement of the movable plate **160** by at least one connector **115** between the first fixed plate **120** and the movable plate **160**. The at least one connector **115** is configured to move the movable plate **160** upon rotation of the actuator ring **190**. The at least one connector **115** can be connected to the movable plate **160** by any suitable connection known to the skilled artisan. For example, as shown in the Figures, the at least one connector **115** comprises a threaded rod connected to the movable plate **160** by a pair of nuts or through a threaded opening in the movable plate **160**. Rotation of the at least two rotary elements **110** causes rotation of the at least one connector **115** around the rotary element axis **113** and movement of the movable plate **160** by interaction of the at least one connector **115** with the movable plate connector **116**. In some embodiments, the movable plate connector **116** comprises one or more nut with threads configured to complement threads on the at least one connector **115**. In some embodiments, the movable plate connector **116** comprises an opening in the movable plate **160** having an internally threaded surface.

(36) The variable orifice valve **100** of some embodiments further comprises one or more of a first conduit **107** extending from first side **122** of the first fixed plate **120**, or a second conduit **108** extending from the second side **144** of the second fixed plate **140**. The first conduit **107** and/or second conduit **108** of some embodiments comprises a suitable connector configured to attach the variable orifice valve **100** to a flow path. Suitable connectors include, but are not limited to, screw threads, compression fittings, direct soldering to the tube, direct lock fittings, and push-connect

fittings (e.g., Shark Bite style fittings). In some embodiments, each of the first conduit **107** and the second conduit **108** is configured to be connected to and form a fluid tight seal with a gas line.

(37) In some embodiments, as shown in FIG. **4**, one or more of the first conduit **107** comprises outwardly facing engagement elements **109** configured to cooperatively interact with the outwardly directed engagement elements **112** of the at least two rotary elements **110**. The engagement elements **109** of some embodiments provides resistance to lateral movement of the at least two rotary elements **110** so that the plurality of outwardly directed engagement elements **112** on the outside surface **111** of the at least two rotary elements **110** remains in contact with the plurality of inwardly directed engagement elements **192** on the inside surface **191** of the actuator ring **190**.

(38) Referring to FIGS. **3**, **4** and **6**, in some embodiments, the needle end **172** of the needle **170** is shaped to cooperatively interact with the opening **126** in the first fixed plate **120**. In some embodiments, the opening **126** of the first fixed plate **120** is sized larger than the needle end **172** so that the entire needle end **172** fits through the **126** without contacting the sides of the opening **126**, as illustrated in FIG. **2**. In embodiments of this sort, the actuator plate **195** has an opening **196** with a maximum diameter that is smaller than the maximum width $W_{sub.T}$ of the needle end **172**. In some embodiments, the maximum width $W_{sub.T}$ of the needle end **172** is sized to contact the second side **124** of the first fixed plate **120** at the opening **126** through the first fixed plate **120** to form a seal.

(39) As shown in FIGS. **6A** and **6B**, in some embodiments, the opening **126** of the first fixed plate **120** is larger than the width $W_{sub.T}$ of the needle end **172** so that actuation of the movable plate **160** causes the needle end **172** to pass through the opening **126** in the first fixed plate **120** without contacting the first fixed plate **120**. The needle end **172** of embodiments of this sort contact the second side **194** of the actuator ring **190** while passing through the opening **196** in the actuator plate **195** to form a fluid-tight seal. In embodiments where the opening **126** of the first fixed plate **120** is greater than the width $W_{sub.T}$ of the needle end **172**, the variable orifice valve **100** conductance can be tuned by replacing the actuator plate **195** with an actuator plate **195** with a different diameter opening **196**.

(40) In use, the needle **170** attached to the movable plate **160** can be moved between a fully opened, see FIG. **6B**, and a fully sealed, see FIG. **6A**, position. This full actuation is referred to as the stroke of the valve. In the embodiment shown, when fully sealed, a portion of the surface of the needle end **172** contacts the second side **194** of the actuator ring **190** actuator plate **195**. Specifically, the needle end **172** contacts the second side **194** at the opening **196**. When the variable orifice valve **100** is opened, the needle **170** backs away from the second side **194** of the actuator plate **195** allowing fluid (e.g., gas) to flow through the opening **196**. The further the needle **170** retracts from the actuator plate **195**, the greater the flow conductance through the variable orifice valve **100**.

(41) In some embodiments, the needle end **172** is conically shaped. The conical shaped needle end **172** can have a sharp end, as shown in FIGS. **6A** and **6B**, or can have a frustoconical end as shown in FIG. **3**, or a rounded conical end, as shown in FIG. **4**. Needle ends **172** having other shapes can also be used without deviating from the scope of the disclosure.

(42) In some embodiments, the needle end **172** has a conical shape with a linear taper, as shown in FIGS. **3**, **4**, **6A** and **6B**. The linear taper can have a truncated, rounded, or sharp end. The angle Θ of the taper of the needle end **172** can be any suitable angle and can affect the conductance of the variable orifice valve **100**. For example, a smaller angle results in a sharper shaped needle end **172** which can provide a longer stroke between a fully opened and sealed valve. Additionally, the sharpness of the needle end **172** can affect the overall conductance of the valve by changing the flow resistance through the valve **100**.

(43) In some embodiments, the needle end **172** has a non-linear taper. For example, the embodiment illustrated in FIG. **7A** has a convex taper and the embodiment illustrated in FIG. **7B** has a concave taper. The skilled artisan will understand the relationship between the shape of the

needle end **172** and the stroke length between fully opened and a fully closed valve.

(44) In some embodiments, the actuator ring **190** has knurling **197** on the outside surface **198**, as shown in FIG. **1**. The knurling **197** can be any suitable pattern or configuration to provide a better grip for the user to rotate the actuator ring **190**. In some embodiments, as shown in FIG. **5**, the actuator ring **190** has marking indicia **199** on the outside surface **198** or top surface (first side **193**) thereof. The marking indicia **199** of some embodiments indicates an amount that the valve **100** is open during rotation of the actuator ring **190**.

(45) FIG. **8** shows an embodiment of the variable orifice valve **100** that is motorized. The embodiment shown comprises a motor **150** that is configured to rotate the actuator ring **190** to rotate the actuator ring **190** around the central axis **105**. The motor **150** can be any suitable motor known to the skilled artisan that will provide sufficient control of the actuator ring **190** and the variable orifice valve **100**. For example, a stepper motor or servo motor can be used.

(46) The motor **150** illustrated in FIG. **8** is connected to the actuator ring **190** using a belt **152** driven by the motor **150**. The belt **152** can be any suitable belt known to the skilled artisan for this purposes. In some embodiments, the belt **152** comprises inwardly facing engagement elements **154** that are configured to cooperatively interact with outwardly facing engagement elements **156** on the actuator ring **190** and outwardly facing engagement elements **155** on the motor pulley **157**. In some embodiments, the belt **152** can be a v-belt that interacts with a complementary groove in the actuator ring **190**.

(47) In some embodiments, the motor **150** is connected to the actuator ring **190** directly, without use of a belt-drive system. In embodiments of this sort, the motor is configured to directly interact with the outwardly facing engagement elements **156** of the actuator ring **190**.

(48) Reference throughout this specification to “one embodiment,” “certain embodiments,” “one or more embodiments” or “an embodiment” means that a particular feature, structure, material, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrases such as “in one or more embodiments,” “in certain embodiments,” “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure.

Furthermore, the particular features, structures, materials, or characteristics may be combined in any suitable manner in one or more embodiments.

(49) Although the disclosure herein has been described with reference to particular embodiments, those skilled in the art will understand that the embodiments described are merely illustrative of the principles and applications of the present disclosure. It will be apparent to those skilled in the art that various modifications and variations can be made to the method and apparatus of the present disclosure without departing from the spirit and scope of the disclosure. Thus, the present disclosure can include modifications and variations that are within the scope of the appended claims and their equivalents.

Claims

1. A variable orifice valve comprising: a first fixed plate at a first end of the variable orifice valve, the first fixed plate having a first side and a second side defining a first fixed plate thickness, the first fixed plate having an opening extending through the first fixed plate thickness; a second fixed plate at a second end of the variable orifice valve, the second fixed plate having a first side and a second side defining a second fixed plate thickness, the second fixed plate having an opening extending through the second fixed plate thickness, the first side of the second fixed plate facing and spaced from the second side of the first fixed plate; a movable plate positioned between the first fixed plate and the second fixed plate, the movable plate having a first side and a second side defining a movable plate thickness, the first side of the movable plate facing the second side of the first fixed plate and the second side of the movable plate facing the first side of the second fixed

plate, the movable plate having a needle extending from the first side of the movable plate, the needle having a needle end sized to fit within the opening in the first fixed plate, the movable plate having at least one opening extending through the movable plate thickness adjacent the needle; a first sealing element between the first fixed plate and the movable plate; a second sealing element between the second fixed plate and the movable plate; an actuator ring adjacent the first fixed plate, the actuator ring having a plurality of inwardly directed engagement elements on an inside surface thereof; at least two rotary elements within the actuator ring, each of the rotary elements having a plurality of outwardly directed engagement elements on an outside surface thereof, each of the outwardly directed engagement elements configured to cooperatively interact with the inwardly directed engagement elements of the actuator ring; and at least one connector between the first fixed plate and the movable plate, the at least one connector configured to move the movable plate upon rotation of the actuator ring.

2. The variable orifice valve of claim 1, further comprising a first conduit extending from the first side of the first fixed plate.
3. The variable orifice valve of claim 2, further comprising a second conduit extending from the second side of the second fixed plate.
4. The variable orifice valve of claim 3, wherein each of the first conduit and the second conduit is configured to be connected to and form a fluid tight seal with a gas line.
5. The variable orifice valve of claim 2, wherein the first conduit comprising outwardly facing engagement elements configured to cooperatively interact with the outwardly directed engagement elements of the rotary elements.
6. The variable orifice valve of claim 5, wherein there are three or more rotary elements arranged symmetrically around the first conduit.
7. The variable orifice valve of claim 6, wherein movement of the actuator ring around a central axis causes each of the rotary elements to rotate around a rotary element axis while remaining in a fixed location relative to the first conduit.
8. The variable orifice valve of claim 7, wherein rotation of each rotary element causes a connector to rotate, the connector rotation causing movement of the movable plate.
9. The variable orifice valve of claim 1, wherein each of the first sealing element and the second sealing element comprises a bellows.
10. The variable orifice valve of claim 9, wherein the bellows comprise a high temperature material.
11. The variable orifice valve of claim 10, wherein a maximum width of the needle end is sized to contact the second side of the first fixed plate at the opening through the first fixed plate to form a seal.
12. The variable orifice valve of claim 1, wherein the needle has a conically shaped end.
13. The variable orifice valve of claim 12, wherein the conically shaped end has a linear taper.
14. The variable orifice valve of claim 12, wherein the conically shaped end has a non-linear taper.
15. The variable orifice valve of claim 1, wherein the actuator ring further comprises knurling on an outside surface thereof.
16. The variable orifice valve of claim 1, wherein the actuator ring has marking indicia on an outside surface or top surface thereof, the marking indicia indicating an amount that the variable orifice valve is open during rotation of the actuator ring.
17. The variable orifice valve of claim 1, wherein there are at least six openings in the movable plate arranged around the needle to allow a fluid to pass through the movable plate.
18. The variable orifice valve of claim 1, further comprising a motor connected to the actuator ring, the motor configured to rotate the actuator ring around a central axis.
19. The variable orifice valve of claim 18, wherein the actuator ring is belt driven by the motor.
20. A variable orifice valve comprising: a first fixed plate at a first end of the variable orifice valve, the first fixed plate having a first side and a second side defining a first fixed plate thickness, the

first fixed plate having an opening extending through the first fixed plate thickness; a second fixed plate at a second end of the variable orifice valve, the second fixed plate having a first side and a second side defining a second fixed plate thickness, the second fixed plate having an opening extending through the second fixed plate thickness, the first side of the second fixed plate facing and spaced from the second side of the first fixed plate; a movable plate positioned between the first fixed plate and the second fixed plate, the movable plate having a first side and a second side defining a movable plate thickness, the first side of the movable plate facing the second side of the first fixed plate and the second side of the movable plate facing the first side of the second fixed plate, the movable plate having a needle extending from the first side of the movable plate, the needle having a needle end sized to fit within the opening in the first fixed plate, the movable plate having at least one opening extending through the movable plate thickness adjacent the needle, the needle end having a conical shape with a maximum width sized to contact the second side of the first fixed plate at the opening to form a seal; a first sealing element between the first fixed plate and the movable plate, the first sealing element comprising a bellows; a second sealing element between the second fixed plate and the movable plate, the second sealing element comprising a bellows; an actuator ring adjacent the first fixed plate, the actuator ring having a plurality of inwardly directed engagement elements on an inside surface thereof; at least three rotary elements within the actuator ring, each of the rotary elements having a plurality of outwardly directed engagement elements on an outside surface thereof, each of the outwardly directed engagement elements configured to cooperatively interact with the inwardly directed engagement elements of the actuator ring; at least one connector between the first fixed plate and the movable plate, the at least one connector configured to move the movable plate upon rotation of the actuator ring; a first conduit extending from the first side of the first fixed plate, the first conduit having a threaded end and outwardly facing engagement elements configured to cooperatively interact with the outwardly directed engagement elements of the at least three rotary elements; and a second conduit extending from the second side of the second fixed plate, the second conduit having a threaded end, wherein movement of the actuator ring around a central axis causes each of the rotary elements to rotate around a rotary element axis while remaining in a fixed location relative to the first conduit.
